

Dr. Markus Götz

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Personal Data

- **Birthday:** 08.09.1988
- **Nationality:** german
- **Place of Birth:** Königs Wusterhausen
- **Martial Status:** married, 2 children

Work Experience

- 01/2020 – ... ➤ **Group Leader** *Karlsruhe Institute of Technology*
Helmholtz AI Energy
- 01/2018 – 12/2019 ➤ **Postdoctoral Researcher** *Karlsruhe Institute of Technology*
Project Management “Helmholtz Analytics Framework (HAF)”
- 07/2016 – 12/2016 ➤ **Research Visit** *University of Iceland, Iceland.*
Signal processing group, Prof. Dr. Ólafur Pétur Pálsson
- 04/2014 – 12/2017 ➤ **Doctoral Studies** *Forschungszentrum Jülich*
Method development for AI applications on high-performance computers.
- 03/2012 – 03/2015 ➤ **Software Developer** *Hasso-Plattner-Institute*
ORE Diagram Editor for Dependability Diagrams with Fuzzy Logic
- 03/2009 – 08/2010 ➤ **Junior Software Developer** *mental Images GmbH*
3D Ray tracing and shader programming

Education

- 04/2014 – 12/2017 ➤ **Ph.D. in Computational Engineering** *University of Iceland*
Thesis title: “Scalable Data Analytics in High Performance Computing”,
Collaboration with the Jülich Supercomputing Centre (JSC),
Examiners: Shantenu Jha, Rutgers University and Håkan Grahm, BTH
- 09/2010 – 03/2014 ➤ **Master of Science in IT-Systems Engineering** *University of Potsdam, Germany*
Grade: A, thesis title: “PyMP – Distributed OpenMP in Python”
- 06/2011 – 09/2011 ➤ **Student** *CERN – European Organization for Nuclear Research, Switzerland*
Analysis of LHCb- and ATLAS collision data
- 08/2010 – 06/2011 ➤ **ERASMUS** *Blekinge Tekniska Högskola, Sweden*
Focus topics: artificial intelligence, parallel programming
- 09/2007 – 07/2010 ➤ **Bachelor of Science in IT-Systems Engineering** *University of Potsdam*
Grade: A, thesis title: “Distributed Data Storages for Google Wave”

Grants

- 2025 > **Carl Zeiss Breakthroughs – WoW**, *Principal Investigator*
Design of joint latent representations for world models, *amount: 750,000€ of 6,000,000€ total*
- > **Helmholtz Association – SLRE**, *Principal Investigator*
Couples AI models for hydrological surrogates, *amount: 325,000€ of 2,500,000€ total*
- 2024 > **Helmholtz Project Call – ARTIST**, *Principal Investigator*
Operational Optimization of Concentrating Solar Power Plants, *amount: 400,000€*
- 2022 > **EU Horizon Project – ASSAS**, *Principal Investigator*
Surrogate model for simulations of severe accidents, *amount: 355,500€*
- 2021 > **HiRSE**, *Principal Investigator*
Research Software Engineering for distributed tensor-tensor computation, *amount: 120,000€ p.a., indefinite base funding*
- > **Helmholtz Programm – Engineering Digital Futures**, *Principal Investigator*
Attack vectors for AI-steered micro grids, *amount: 97,000€*
- > **Helmholtz Project Call – AI InsuPero**, *Principal Investigator*
Optimization of solar cell manufacturing with AI, *amount: 400,000€*
- 2020 > **Helmholtz Project Call – Folding by Learning**, *Principal Investigator*
Prediction of RNA contacts with self-supervised models, *amount: 370,000€.*
- 2019 > **Helmholtz AI Local Energy**, *Principle investigator*
AI research and consulting group, *amount: 1,050,000€ p.a., indefinite base funding*

List of Publications

Peer-reviewed Conference Proceedings

- [1] Huber, Philipp, David Li, Juan Pedro Gutiérrez Hermosillo Muriedas, Deifilia Kieckhefen, Markus Götz, Achim Streit, and Charlotte Debus (2026). “Energy Consumption in Parallel Neural Network Training”. In: *Proceedings of the 2026 SIAM Conference on Parallel Processing for Scientific Computing (PP)*. SIAM, pp. 46–59. DOI: 10.1137/1.9781611979022.4.
- [2] Karolin Özdemir, Asena, Lars H Heyen, Arvid Weyrauch, Achim Streit, Markus Götz, and Charlotte Debus (2026). “Sampling Parallelism for Fast and Efficient Bayesian Learning”. In: *Proceedings of the Platform for Advanced Scientific Computing Conference (PASC ’26)*.
- [3] Kieckhefen, Deifilia, Markus Götz, Lars H Heyen, Achim Streit, and Charlotte Debus (2026). “Jigsaw: Training Multi-Billion-Parameter AI Weather Models with Optimized Model Parallelism”. In: *Proceedings of the Platform for Advanced Scientific Computing Conference (PASC ’26)*.
- [4] Flügel, Katharina, Daniel Coquelin, Marie Weiel, Charlotte Debus, Achim Streit, and Markus Götz (2025). “Beyond Backpropagation: Optimization with Multi-Tangent Forward Gradients”. In: *2025 International Joint Conference on Neural Networks (IJCNN)*. IEEE, pp. 1–8. DOI: 10.1109/IJCNN64981.2025.11227446.
- [5] Herold, Julian, Paul Brinkmeier, Markus Götz, Alexander Schug, and Achim Streit (2025). “Large-Scale Modeling of the Extracellular Matrix on Hybrid Supercomputers Using Highly Parallel Stencil Computations”. In: *ISC High Performance 2025 Research Paper Proceedings (40th International Conference)*. Prometheus GmbH, pp. 1–14.
- [6] Kuhn, Nicholas, Arvid Weyrauch, Muhamed Öz, Achim Streit, Markus Götz, and Charlotte Debus (2025). “A Comparative Study of Pruning Methods in Transformer-Based Time Series Forecasting”. In:

2025 IEEE International Conference on Data Mining Workshops (ICDMW). IEEE, pp. 223–228.  DOI: 10.1109/ICDMW69685.2025.00032.

- [7] Tuteja, Keshvi, Gregor Olenik, Roman Mishchuk, Yu-Hsiang Tsai, Markus Götz, Achim Streit, Hartwig Anzt, and Charlotte Debus (2025). “pyGinkgo: A Sparse Linear Algebra Operator Framework for Python”. In: *Proceedings of the 54th International Conference on Parallel Processing*, pp. 753–763.  DOI: 10.1145/3754598.3754648.
- [8] Coquelin, Daniel, Katharina Flügel, Marie Weiel, Nicholas Kiefer, Charlotte Debus, Achim Streit, and Markus Götz (2024). “Harnessing Orthogonality to Train Low-Rank Neural Networks”. In: *27th European Conference on Artificial Intelligence (ECAI 2024)*. 392, pp. 2106–2113.  DOI: 10.3233/FAIA240729.
- [9] Debus, Jan, Charlotte Debus, Günther Dissertori, and Markus Götz (2024). “PETNet – Coincident Particle Event Detection using Spiking Neural Networks”. In: *Proceedings of the 2024 Annual Neuro-Inspired Computational Elements Conference*. ACM.  DOI: 10.1109/NICE61972.2024.10549584.
- [10] Flügel, Katharina, Daniel Coquelin, Marie Weiel, Charlotte Debus, Achim Streit, and Markus Götz (2024). “Feed-Forward Optimization With Delayed Feedback”. In: *International Conference on Neural Information Processing 2024 (ICONIP 2024)*.
- [11] Öz, Muhammed, Nicholas Kiefer, Charlotte Debus, Jasmin Hörter, Achim Streit, and Markus Götz (2024). “Model Fusion via Neuron Transplantation”. In: *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*. Springer, pp. 3–19.  DOI: 10.1007/978-3-031-70359-1_1.
- [12] Pütz, Sebastian, Hadeer El Ashhab, Matthias Hertel, Ralf Mikut, Markus Götz, Veit Hagenmeyer, and Benjamin Schäfer (2024). “Feasibility of Forecasting Highly Resolved Power Grid Frequency Utilizing Temporal Fusion Transformers”. In: *Proceedings of the 15th ACM International Conference on Future and Sustainable Energy Systems*, pp. 447–453.  DOI: 10.1145/3632775.3661963.
- [13] Weyrauch, Arvid, Thomas Steens, Oskar Taubert, Benedikt Hanke, Aslan Egbal, Ewa Götz, Achim Streit, Markus Götz, and Charlotte Debus (2024). “ReCycle: Fast and Efficient Long Time Series Forecasting with Residual Cyclic Transformers”. In: *Proceedings of the 2024 IEEE Conference on Artificial Intelligence (IEEE CAI)*. IEEE.  DOI: 10.1109/CAI59869.2024.00212.
- [14] Zwerschke, Pavel, Arvid Weyrauch, Markus Götz, and Charlotte Debus (2024). “Taylor Expansion in Neural Networks: How Higher Orders Yield Better Predictions”. In: *27th European Conference on Artificial Intelligence (ECAI 2024)*. 392, pp. 2983–2989.  DOI: 10.3233/FAIA240838.
- [15] Gutiérrez Hermosillo Muriedas, Juan Pedro, Katharina Flügel, Charlotte Debus, Holger Obermaier, Achim Streit, and Markus Götz (2023). “perun: Benchmarking Energy Consumption of High-Performance Computing Applications”. In: *European Conference on Parallel Processing*. Springer, pp. 17–31.  DOI: 10.1007/978-3-031-39698-4_2.
- [16] Klein, Lukas, Sebastian Ziegler, Felix Laufer, Charlotte Debus, Markus Götz, Klaus Maier-Hein, Ulrich Paetzold, Fabian Isensee, and Paul Jaeger (2023). “Understanding Scalable Perovskite Solar Cell Manufacturing with Explainable AI”. In: *XAI in Action: Past, Present, and Future Applications*.
- [17] Taubert, Oskar, Marie Weiel, Daniel Coquelin, Anis Farshian, Charlotte Debus, Alexander Schug, Achim Streit, and Markus Götz (2023). “Massively Parallel Genetic Optimization Through Asynchronous Propagation of Populations”. In: *International Conference on High Performance Computing*. Springer, pp. 106–124.  DOI: 10.1007/978-3-031-32041-5_6.
- [18] Caspart, René, Sebastian Ziegler, Arvid Weyrauch, Holger Obermaier, Simon Raffener, Leon Pascal Schuhmacher, Jan Scholtyssek, Darya Trofimova, Marco Nolden, Ines Reinartz, Fabian Isensee, Markus Götz, and Charlotte Debus (2022). “Precise Energy Consumption Measurements of Heterogeneous Artificial Intelligence Workloads”. In: *International Conference on High Performance Computing*. Springer, pp. 108–121.  DOI: 10.1007/978-3-031-23220-6_8.

- [19] Coquelin, Daniel, Behnood Rasti, Markus Götz, Pedram Ghamisi, Richard Gloaguen, and Achim Streit (2022). "Hyde: The First Open-Source, Python-Based, Gpu-Accelerated Hyperspectral Denoising Package". In: *2022 12th Workshop on Hyperspectral Imaging and Signal Processing: Evolution in Remote Sensing (WHISPERS)*, pp. 1–5.  DOI: 10.1109/WHISPERS56178.2022.9955088.
- [20] Funk, Yannick, Markus Götz, and Hartwig Anzt (2022). "Prediction of Optimal Solvers for Sparse Linear Systems Using Deep Learning". In: *Proceedings of the 2022 SIAM Conference on Parallel Processing for Scientific Computing*. SIAM, pp. 14–24.  DOI: 10.1137/1.9781611977141.2.
- [21] Schilling, Marcel P, Oliver Neumann, Tim Scherr, Haijun Cui, Anna A Popova, Pavel A Levkin, Markus Götz, and Markus Reischl (2022). "A computational workflow for interdisciplinary deep learning projects utilizing bwHPC infrastructure". In: *Proceedings of the 7th bwHPC Symposium*. Vol. 7. Universität Ulm, pp. 69–74.
- [22] Coquelin, Daniel, Rocco Sedona, Morris Riedel, and Markus Götz (2021). "Evolutionary Optimization of Neural Architectures in Remote Sensing Classification Problems". In: *2021 IEEE International Geoscience and Remote Sensing Symposium IGARSS*. IEEE, pp. 1587–1590.  DOI: 10.1109/IGARSS47720.2021.9554309.
- [23] Götz, Markus, Charlotte Debus, Daniel Coquelin, Kai Krajsek, Claudia Comito, Philipp Knechtges, Björn Hagemeyer, Michael Tarnawa, Simon Hanselmann, Martin Siggel, Achim Basermann, and Achim Streit (2020). "HeAT – a Distributed and GPU-accelerated Tensor Framework for Data Analytics". In: *2020 IEEE International Conference on Big Data (Big Data)*, pp. 276–287.  DOI: 10.1109/BigData50022.2020.9378050.
- [24] Taubert, Oskar, Markus Götz, Alexander Schug, and Achim Streit (2020). "Loss Scheduling for Class-Imbalanced Image Segmentation Problems". In: *2020 19th International Conference on Machine Learning and Applications (ICMLA)*. IEEE, pp. 426–431.  DOI: 10.1109/ICMLA51294.2020.00073.
- [25] Cavallaro, Gabriele, Valentin Kozlov, Markus Götz, and Morris Riedel (2019). "Remote Sensing Data Analytics with the UDocker Container Tool using Multi-GPU Deep Learning Systems". In: *Conference on Big Data from Space (BiDS'19)*.  DOI: 10.2760/848593.
- [26] Kleinert, Felix, Bing Gong, Markus Götz, and Martin Georg Schultz (2019). "Near Surface Ozone Predictions Based on Multiple Artificial Neural Network Architectures". In: *Geophysical Research Abstracts*. Vol. 21.  DOI: 2128/23877.
- [27] Götz, Markus and Hartwig Anzt (2018). "Machine Learning-Aided Numerical Linear Algebra: Convolutional Neural Networks for the Efficient Preconditioner Generation". In: *Proceedings of the 9th Workshop on Latest Advances in Scalable Algorithms for Large-Scale Systems (ScalA)*. IEEE, pp. 49–56.  DOI: 10.1109/ScalA.2018.00010.
- [28] Krajsek, Kai, Claudia Comito, Markus Götz, Björn Hagemeyer, Philipp Knechtges, and Martin Siggel (2018). "The Helmholtz Analytics Toolkit (HeAT): a scientific Big Data Library for HPC". In: *Proceedings of the Extreme Data Workshop*.  DOI: 10.5445/IR/1000089169.
- [29] Lange, Julius, Gabriele Cavallaro, Markus Götz, Ernir Erlingsson, and Morris Riedel (2018). "The Influence of Sampling Methods on Pixel-Wise Hyperspectral Image Classification with 3D Convolutional Neural Networks". In: *Proceedings of the 2018 IEEE Geoscience and Remote Sensing Symposium (IGARSS)*. IEEE, pp. 2087–2090.  DOI: 10.1109/IGARSS.2018.8518671.
- [30] Book, Matthias, Morris Riedel, Helmut Neukirchen, and Markus Götz (2017). "Facilitating Collaboration in High-performance Computing Projects with an Interaction Room". In: *Proceedings of the 4th ACM SIGPLAN International Workshop on Software Engineering for Parallel Systems*. ACM, pp. 46–47.  DOI: 10.1145/3141865.3142467.
- [31] Götz, Markus, Matthias Book, Christian Bodenstein, and Morris Riedel (2017). "Supporting Software Engineering Practices in the Development of Data-Intensive HPC Applications with the JuML Framework". In: *Proceedings of the International Workshop on Software Engineering for High Performance*

Computing in Computational and Data-enabled Science & Engineering. ACM, pp. 1–8.  DOI: 10.1145/3144763.3144765.

- [32] Bodenstein, Christian, Markus Götz, Annika Jansen, Henrike Scholz, and Morris Riedel (2016). “Automatic Object Detection Using DBSCAN for Counting Intoxicated Flies in the FLORIDA Assay”. In: *Proceedings of the 15th IEEE International Conference on Machine Learning and Applications (ICMLA)*. IEEE, pp. 746–751.  DOI: 10.1109/ICMLA.2016.0133.
- [33] Cavallaro, Gabriele, Morris Riedel, Christian Bodenstein, Philipp Glock, Matthias Richerzhagen, Markus Götz, and Jon Atli Benediktsson (2015). “Scalable Developments for Big Data Analytics in Remote Sensing”. In: *Proceedings of the 2015 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*. IEEE, pp. 1366–1369.  DOI: 10.1109/IGARSS.2015.7326030.
- [34] Götz, Markus, Christian Bodenstein, and Morris Riedel (2015). “HPDBSCAN: Highly Parallel DBSCAN”. In: *Proceedings of the Workshop on Machine Learning in High-Performance Computing Environments (MLHPC ’15)*, pp. 1–10.  DOI: 10.1145/2834892.2834894.
- [35] Götz, Markus and Morris Riedel (2015). “On Parallel and Scalable Classification and Clustering Techniques for Earth Science Datasets”. In: *Proceedings of the Sixth Workshop on Data Mining in Earth System Science (DMESS)*.
- [36] Riedel, Morris, Markus Götz, Matthias Richerzhagen, Philipp Glock, Christian Bodenstein, Ahmed Shiraz Memon, and M Shahbaz Memon (2015). “Scalable and Parallel Machine Learning Algorithms for Statistical Data Mining—Practice & Experience”. In: *Proceedings of the 38th International Convention on Information and Communication Technology, Electronics and Microelectronics (MIPRO)*. IEEE, pp. 204–209.  DOI: 10.1109/MIPRO.2015.7160265.
- [37] Cavallaro, Gabriele, Morris Riedel, Jon Atli Benediktsson, Markus Götz, Tomas Runarsson, Kristjan Jonasson, and Thomas Lippert (2014). “Smart Data Analytics Methods for Remote Sensing Applications”. In: *Proceedings of the 2014 IEEE Geoscience and Remote Sensing Symposium (IGARSS)*. IEEE, pp. 1405–1408.  DOI: 10.1109/IGARSS.2014.6946698.
- [38] Goderbauer, Michael, Markus Götz, Alexander Grosskopf, Andreas Meyer, and Mathias Weske (2010). “Syncro-Concurrent Editing Library for Google Wave”. In: *Proceedings of the 10th International Conference on Web Engineering (ICWE’10)*. Springer, pp. 510–513.  DOI: 10.1007/978-3-642-13911-6_40.

Journal Papers

- [1] Heuel, Janis, Arnaud Delsuc, Marie Weiel, Daniel Coquelin, Markus Götz, and Andreas Rietbrock (2026). “Optimising Deep-Learning First-Break Picking”. In: *RAS Techniques and Instruments*, rzago27.  DOI: 10.1093/rasti/rzag027.
- [2] Nikoltchovska, Alexandra, Sebastian Pütz, Markus Götz, and Benjamin Schäfer (2026). “Correlations Matter in Explanations for Energy Systems”. In: *KI-Künstliche Intelligenz*, pp. 1–14.  DOI: 10.1007/s13218-026-00904-4.
- [3] Weyrauch, Arvid, Lars H Heyen, Juan Pedro Gutiérrez Hermosillo Muriedas, Pei-Hsuan Hsia, Asena Karolin Özdemir, Achim Streit, Markus Götz, and Charlotte Debus (2026). “torch_blue: A Flexible Python Package for Bayesian Neural Networks in PyTorch”. In: *Journal of Open Source Software* 11.117, p. 9415.  DOI: 10.21105/joss.09415.
- [4] Flügel, Katharina, Charlotte Debus, Markus Götz, Achim Streit, and Marie Weiel (2025). “Scaling Laws of Distributed Random Forests”. In: *Transactions on Machine Learning Research*.
- [5] Gazos, Alexandros, James Kahn, Isabel Kusche, Christian Büscher, and Markus Götz (2025). “Organising AI for safety: Identifying structural vulnerabilities to guide the design of AI-enhanced socio-technical systems”. In: *Safety Science* 184, p. 106731.  DOI: 10.1016/j.ssci.2024.106731.

- [6] Götz, Markus, Erik Wessel, Uğur Çayoğlu, Babak Jahani, Charlotte Debus, Jan Cermak, and Achim Streit (2025). "Ordinal METAR Cloud Cover Classification in Sentinel-2 Satellite Data". In: *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*. DOI: 10.1109/JSTARS.2025.3557931.
- [7] Hoppe, Fabian, Juan Pedro Gutiérrez Hermosillo Muriedas, Michael Tarnawa, Philipp Knechtges, Björn Hagemeier, Kai Krajsek, Alexander Rüttgers, Markus Götz, and Claudia Comito (2025). "Engineering a large-scale data analytics and array computing library for research: Heat". In: *Electronic Communications of the EASST* 83.
- [8] Huang, Yu-Chieh, Pierre Tremouilhac, Stefan Kuhn, Pei-Chi Huang, Chia-Lin Lin, Nils Schlörer, Oskar Taubert, Markus Götz, Nicole Jung, and Stefan Bräse (2025). "(Semi-) Automatic Review Process for Common Compound Characterization Data in Organic Synthesis". In: *arxiv*.
- [9] Laufer, Felix, Markus Götz, and Ulrich Wilhelm Paetzold (2025). "Deep learning for augmented process monitoring of scalable perovskite thin-film fabrication". In: *Energy & Environmental Science*. DOI: 10.1039/D4EE03445G.
- [10] Meisenbacher, Stefan, Kaleb Phipps, Oskar Taubert, Marie Weiel, Markus Götz, Ralf Mikut, and Veit Hagenmeyer (2025). "AutoPQ: Automating quantile estimation from point forecasts in the context of sustainability". In: *Applied Energy* 392, p. 125931. DOI: 10.1016/j.apenergy.2025.125931.
- [11] Phipps, Kaleb, Mathias Kuhl, Marie Weiel, Marlene Busch, Jan Lewen, Nicolas Blumenröhr, Daniel Maldonado Quinto, Charlotte Debus, Felix Göhring, Oliver Kaufhold, et al. (2025). "PAINT: The First FAIR Database for Concentrating Solar Power Plants". In: *Nature Energy*.
- [12] Upadhyay, Utkarsh, Julian Herold, Markus Götz, and Alexander Schug (2025). "NucleicBERT: Deciphering the language of nucleic acids by a large-language model". In: *bioRxiv*, pp. 2025–09. DOI: 10.1101/2025.09.02.673754.
- [13] Vollmer, Elena, Mishal Benz, James Kahn, Rebekka Volk, Fran Schultmann, and Markus Götz (2025). "Enhancing UAV-based Multispectral Semantic Segmentation through Feature Engineering - An RGB and Thermal Image Ablation Study". In: *Journal of Selected Topics in Applied Earth Observations and Remote Sensing*.
- [14] Flügel, Katharina, Daniel Coquelin, Marie Weiel, Achim Streit, and Markus Götz (2024). "Beyond Backpropagation: Optimization with Multi-Tangent Forward Gradients". In: *arXiv preprint arXiv:2410.17764*.
- [15] Pargmann, Max, Jan Ebert, Markus Götz, Daniel Maldonado Quinto, Robert Pitz-Paal, and Stefan Kesselheim (2024). "Automatic heliostat learning for in situ concentrating solar power plant metrology with differentiable ray tracing". In: *Nature Communications* 15.1, p. 6997. DOI: 10.1038/s41467-024-51019-z.
- [16] To, Deifilia, Julian Quinting, Gholam Ali Hoshyaripour, Markus Götz, Achim Streit, and Charlotte Debus (2024). "Architectural insights into and training methodology optimization of Pangu-Weather". In: *Geoscientific Model Development* 17.23, pp. 8873–8884. DOI: 10.5194/egusphere-2024-1714.
- [17] Xavier, Joseph Arnold, Juan Pedro Gutiérrez Hermosillo Muriedas, Stepan Nassyr, Rocco Sedona, Markus Götz, Achim Streit, Morris Riedel, and Gabriele Cavallaro (2024). "Vectorized Highly Parallel Density-based Clustering for Applications with Noise". In: *IEEE Access*. DOI: 10.1109/ACCESS.2024.3507193.
- [18] Debus, Charlotte, Marie Piraud, Achim Streit, Fabian Theis, and Markus Götz (2023). "Reporting electricity consumption is essential for sustainable AI". In: *Nature Machine Intelligence*, pp. 1–3. DOI: 10.1038/s42256-023-00750-1.

- [19] Farshian, Anis, Markus Götz, Gabriele Cavallaro, Charlotte Debus, Matthias Nießner, Jón Atli Benediktsson, and Achim Streit (2023). “Deep-Learning-Based 3-D Surface Reconstruction—A Survey”. In: *Proceedings of the IEEE*. DOI: 10.1109/JPROC.2023.3321433.
- [20] Klein, Lukas, Sebastian Ziegler, Felix Laufer, Charlotte Debus, Markus Götz, Klaus Maier-Hein, Ulrich W Paetzold, Fabian Isensee, and Paul F Jäger (2023). “Discovering Process Dynamics for Scalable Perovskite Solar Cell Manufacturing with Explainable AI”. In: *Advanced Materials*, p. 2307160. DOI: 10.1002/adma.202307160.
- [21] Laufer, Felix, Sebastian Ziegler, Fabian Schackmar, Edwin A Moreno Viteri, Markus Götz, Charlotte Debus, Fabian Isensee, and Ulrich W Paetzold (2023). “Process Insights into Perovskite Thin-Film Photovoltaics from Machine Learning with In Situ Luminescence Data”. In: *Solar RRL*. DOI: 10.1002/solr.202201114.
- [22] Mayer, Zoe, James Kahn, Markus Götz, Yu Hou, Tobias Beiersdörfer, Nicolas Blumenröhr, Rebekka Volk, Achim Streit, and Frank Schultmann (2023). “Thermal Bridges on Building Rooftops”. In: *Scientific Data* 10.1, p. 268. DOI: 10.1038/s41597-023-02140-z.
- [23] Mayer, Zoe, James Kahn, Yu Hou, Markus Götz, Rebekka Volk, and Frank Schultmann (2023). “Deep learning approaches to building rooftop thermal bridge detection from aerial images”. In: *Automation in Construction* 146. DOI: 10.1016/j.autcon.2022.104690.
- [24] Piraud, Marie, Andrés Camero, Markus Götz, Stefan Kesselheim, Peter Steinbach, and Tobias Weigel (2023). “Providing AI expertise as an infrastructure in academia”. In: *Patterns* 4.8. DOI: 10.1016/j.patter.2023.100819.
- [25] Taubert, Oskar, Fabrice von der Lehr, Alina Bazarova, Christian Faber, Philipp Knechtges, Marie Weiel, Charlotte Debus, Daniel Coquelin, Achim Basermann, Achim Streit, Stefan Kesselheim, Markus Götz, and Alexander Schug (2023). “RNA Contact Prediction by Data Efficient Deep Learning”. In: *Communications Biology* 6 (1). DOI: 10.1038/s42003-023-05244-9.
- [26] Coquelin, Daniel, Charlotte Debus, Markus Götz, Fabrice von der Lehr, James Kahn, Martin Siggel, and Achim Streit (2022). “Accelerating neural network training with distributed asynchronous and selective optimization (DASO)”. In: *Journal of Big Data* 9.1, pp. 1–18. DOI: 10.1186/s40537-021-00556-1.
- [27] Kahn, James, Ilias Tsaklidis, Oskar Taubert, Lea Reuter, Giulio Dujany, Tobias Boeckh, Arthur Thaller, Pablo Goldenzweig, Florian Bernlochner, Achim Streit, et al. (2022). “Learning tree structures from leaves for particle decay reconstruction”. In: *Machine Learning: Science and Technology* 3.3, p. 035012. DOI: 10.1088/2632-2153/ac8de0.
- [28] Weiel, Marie, Markus Götz, André Klein, Daniel Coquelin, Ralf Floca, and Alexander Schug (2021). “Dynamic particle swarm optimization of biomolecular simulation parameters with flexible objective functions”. In: *Nature Machine Intelligence* 3.8, pp. 727–734. DOI: 10.1038/s42256-021-00366-3.
- [29] Götz, Markus, Mikhail Kononets, Christian Bodenstein, Morris Riedel, Matthias Book, and Olafur Petur Palsson (2019). “Automatic Water Mixing Event Identification in the Koljö Fjord Observatory Data”. In: *International Journal of Data Science and Analytics* 7.1, pp. 67–79. DOI: 10.1007/s41060-018-0132-z.
- [30] Götz, Markus, Gabriele Cavallaro, Thierry Geraud, Matthias Book, and Morris Riedel (2018). “Parallel Computation of Component Trees on Distributed Memory Machines”. In: *IEEE Transactions on Parallel and Distributed Systems* 29.11, pp. 2582–2598. DOI: 10.1109/TPDS.2018.2829724.
- [31] Götz, Markus, Matthias Richerzhagen, Christian Bodenstein, Gabriele Cavallaro, Philipp Glock, Morris Riedel, and Jón Atli Benediktsson (2015). “On Scalable Data Mining Techniques for Earth Science”. In: *Procedia Computer Science* 51, pp. 2188–2197. DOI: 10.1016/j.procs.2015.05.494.

Invited Talks

- [1] Götz, Markus (2024a). “Karlsruhe Adaptive Systems Talks Key Note”. In: TRIANGEL Karlsruhe.

- [2] Götz, Markus (2024b). "STULLE - wissenschaftlich beleg". In: TRIANGEL Karlsruhe.
- [3] — (2024c). "Wärmebrückenerkennung in Drohnenbildern mit Deep Learning". In: Leipzig Data Week.
- [4] — (2022). "Towards Sustainable AI in Energy Research". In: H.AI Conference 2022.
- [5] Götz, Markus and Peter Steinbach (2022). "Testing Neural Applications". In: GI Software Engineering 2022 (SE22).
- [6] Götz, Markus (2021). "AI in Energy Research". In: KCIST Workshop.
- [7] — (2020). "HeAT – a Distributed and GPU-accelerated Tensor Framework." In: Spanish Supercomputing Network (RES), Barcelona Supercomputing Centre (BSC).
- [8] — (2019a). "A Compute Center Perspective on Big Data". In: GI-Dagstuhl Seminar Algorithms for Big Data.
- [9] — (2019b). "Machine Learning with Neural Networks". In: GridKa Summer School.
- [10] — (2018a). "Cluster Analysis". In: 1st DAME Workshop, Hamburg.
- [11] — (2018b). "Machine Learning with Neural Networks". In: GridKa Summer School.
- [12] — (2015). "Data Analytics on HPC". In: 3rd Joint Laboratory for Extreme-Scale Computing (JLESC) Summer School, Barcelona.

Datasets

- 2025 > **METAR** – Cloud cover above weather stations mapped to Sentinel-2 satellite images, 4 contributors
<https://doi.org/10.5281/zenodo.14691473>
- 2024 > **PAINT** – Operational data of a concentrating solar power plant, 13 contributors
<https://paint-database.org/>
- 2022 > **TBBR** – Thermal Bridges on Building Rooftops - Hyperspectral (RGB + Thermal + Height) drone images of Karlsruhe, Germany, with thermal bridge annotations, 6 contributors
<https://doi.org/10.5281/zenodo.7022736>
- > **LCAG** – Lowest Common Ancestor Generations (LCAG) Phasespace Particle Decay Reconstruction Dataset, 11 contributors
<https://doi.org/10.5281/zenodo.6983258>

Software

- 2024 – ... > **ARTIST**
 AI-guided differentiable ray tracer for concentrating solar power plants, 6 contributors, <https://github.com/ARTIST-Association/ARTIST>
- > **PAINT**
 Source code for the PAINT dataset, 6 contributors, <https://github.com/ARTIST-Association/PAINT>
- > **CANVAS**
 3D Editor for ARTIST, 9 contributors, <https://github.com/ARTIST-Association/CANVAS>
- 2023 – ... > **perun**
 Energy consumption measurements for parallel Python application, 5 contributors
<https://github.com/Helmholtz-AI-Energy/perun>

Software (cont.)

- 2022 – . . . > **Propulate**
Massively parallel and asynchronous evolutionary optimizer, 6 contributors
<https://github.com/Helmholtz-AI-Energy/propulate>
- 2021 – . . . > **HyDE – Hyperspectral Denoising**
Library for GPU-accelerated neural image denoising, 2 contributors
<https://github.com/Helmholtz-AI-Energy/HyDe>
- 2018 – . . . > **HeAT – Helmholtz Analytics Toolkit**
Distributed tensor framework for high-performance AI, 32 contributors
<https://github.com/helmholtz-analytics/heat>
- 2017 – . . . > **Distributed Max Trees (DMT)**
Computation of max trees on distributed memory machines, 2 contributors
<https://github.com/Markus-Goetz/dmt>
- 2015 – . . . > **Highly Parallel DBSCAN (HPDBSCAN)**
Parallel Distributed-memory DBSCAN Algorithm, 6 contributors
<https://github.com/Markus-Goetz/hpdbscan>
- 2012 – 2015 > **Open Reliability Editor (ORE)**
ORE Diagram Editor for Dependability Diagrams, 9 contributors
<https://github.com/troeeger/fuzzed>

Prices

Awards

- 2025 > **Best Poster Award**
“PAINT: A FAIR Database for Concentrating Solar Power Plants”, HAICON 2025
- > **Best Poster Award**
“pyGinko: A Sparse Linear Algebra Operator Framework for Python”, ISC 2025
- 2024 > **Best Paper Award**
“Feed-Forward Optimization With Delayed Feedback for Neural Networks”, ICONIP 2024
- 2018 > **Best Tutorial GridKa School**
“Machine Learning with Neural Networks”
- 2016 > **Best Poster Award IAS Symposium,**
Forschungszentrum Jülich
- 2014 > **Distinction Medal Master Program,** *University of Potsdam*
- 2010 > **Google- und Axel-Springer-Scholarship for Bachelor Thesis**
Presentation of bachelor project on Google I/O 2010

Teaching Experience

- WT 2024/25 > **Scalable AI, Karlsruhe Institute of Technology**
Teaching assignment by Department of Informatics at KIT, 4 ECTS points
- WT 2023/24 > **Scalable AI, Karlsruhe Institute of Technology**
Teaching assignment by Department of Informatics at KIT, 4 ECTS points
- WT 2022/23 > **Scalable AI, Karlsruhe Institute of Technology**
Teaching assignment by Department of Informatics at KIT, 4 ECTS points

Teaching Experience (cont.)

- WT 2021/22 > **Scalable AI**, *Karlsruhe Institute of Technology*
Teaching assignment by Department of Informatics at KIT, 4 ECTS points
- WT 2020/21 > **High Performance Computing**, *Karlsruhe Institute of Technology*
Instructor, 3 ECTS points, *englisch*, block lecture
- ST 2020 > **Advanced Topics in Parallel Programming**, *Karlsruhe Institute of Technology*
Seminar, Data-parallel Training of Neural Networks, *englisch*, 6 ECTS points.
- ST 2019 > **Machine Learning For Automated Linear Solver Selection**, *KIT*
Group seminar, jointly with Prof. Hartwig Anzt, 6 ECTS points
- > **Advanced Topics in Parallel Programming**, *Karlsruhe Institute of Technology*
Seminar, Data-parallel Training of Neural Networks, *englisch*, 6 ECTS points.
- WT 2018/19 > **Software Engineering Lab**, *Karlsruhe Institute of Technology*
Seminar, 6 ECTS points
- WT 2017/18 > **High-Performance Computing**, *University of Iceland*
Co-instructor, five lectures, *english*, 7.5 ECTS points
- > **Statistical Data Mining**, *University of Iceland*
Co-instructor, two lectures, *englisch*, 7.5 ECTS points
- WT 2016/17 > **Distributed Systems**, *University of Iceland*
Co-instructor, four lectures, *englisch*, 7.5 ECTS points
- ST 2016 > **Parallelization of Distributed CART Decision Trees**, *RWTH Aachen*
Seminar, 6 ECTS points

Supervision

Doctoral Studies

- 01/2024 – ... > **Optimal Sparse Computational Kernels for Neural Networks**, *Karlsruhe Institute of Technology*, Keshvi Tuteja, main supervisor: Prof. Dr. Achim Streit
- 12/2023 – ... > **Optimizing Electric Grids with Data-driven Approaches**, *Karlsruhe Institute of Technology*, Muhammed Öz, main supervisor: Prof. Dr. Achim Streit
- 05/2022 – ... > **Model-parallel Neural Networks**, *Karlsruhe Institute of Technology*, Katharina Flügel, main supervisor: Prof. Dr. Achim Streit
- 04/2022 – ... > **Resource-optimal Distributed Tensors**, *Karlsruhe Institute of Technology*, Juan Pedro Gutierrez, main supervisor: Prof. Dr. Achim Streit
- 07/2021 – 10/2024 > **Data-parallel Neural Networks**, *Karlsruhe Institute of Technology*, Daniel Coquelin, main supervisor: Prof. Dr. Achim Streit
- 08/2020 – ... > **Deep Learning for Adaptive Biological Simulation Control**, *Karlsruhe Institute of Technology*, Julian Herold, main supervisor: Prof. Dr. Alexander Schug.
- 04/2019 – 02/2024 > **Scalable AI-driven Modeling of Biomolecules**, *Karlsruhe Institute of Technology*, Oskar Taubert, main supervisor: Prof. Dr. Achim Streit
- 04/2018 – 08/2021 > **Deriving protein structures efficiently by integrating experimental data into biomolecular simulations**, *Karlsruhe Institute of Technology*, Marie Weiel, main supervisor: Prof. Dr. Alexander Schug

Master Thesis

- 01/2026 – ... > **Adaptive Runtime Optimizations for MPI Communication of Non-Contiguous Tensor Data Using Decision Trees**, *Karlsruhe Institute of Technology*, Vito Dierksen, main supervisor: Prof. Dr. Achim Streit

Supervision (cont.)

- | | |
|-------------------|---|
| 05/2025 – 11/2025 | > Enhanced Tangent-Selection for Forward-Gradients in Neural Networks , <i>Karlsruhe Institute of Technology</i> , Tobias Paweletz, main supervisor: Prof. Dr. Achim Streit |
| | > Forward Learning Pipeline Parallelism for Deep Neural Networks , <i>Karlsruhe Institute of Technology</i> , Joshua Hauth, main supervisor: Prof. Dr. Achim Streit |
| 04/2025 – 10/2025 | > Optimisation of Distributed Tensor Applications Using Reinforcement Learning (Optimisation of Distributed Tensor Applications Using Reinforcement Learning) , <i>Karlsruhe Institute of Technology</i> , Vadim Kraynyukov, main supervisor: Prof. Dr. Achim Streit |
| 04/2024 – 10/2024 | > Evaluation of AI-driven weather forecast adaption for the wind energy sector , <i>Karlsruhe Institute of Technology</i> , Elias König, main supervisor: Prof. Dr. Melanie Schienle |
| | > Post-hoc Explanations for Probabilistic Time Series Forecasts , Yannick Funk, main supervisor: Prof. Dr. Achim Streit |
| 06/2023 – 11/2023 | > Parallel Neural Transplantation , <i>Karlsruhe Institute of Technology</i> , Muhammed Öz, main supervisor: Prof. Dr. Achim Streit |
| 07/2022 – 02/2023 | > Predicting ILU(o) Effectiveness for Sparse Matrix Systems via Explainable Machine learning , <i>Karlsruhe Institute of Technology</i> , Anna Ostrovskaya, main supervisor: Prof. Dr. Hartwig Anzt |
| 06/2021 – 12/2021 | > Feed-forward-only Training of Neural Networks , <i>Karlsruhe Institute of Technology</i> , Katharina Flügel, main supervisor: Prof. Dr. Achim Streit |
| 01/2019 – 06/2019 | > Demonstrating Learning Particle Decay Reconstruction Using Graph Neural Networks at Belle II , <i>University of Strasbourg, Karlsruhe Institute of Technology</i> , Ilias Tsaklidis, main supervisor: Prof. Dr. Isabelle Ripp-Baudot |
| 07/2018 – 01/2019 | > Accelerating Molecular Dynamics Simulations using Convolutional Variational Autoencoders , <i>Karlsruhe Institute of Technology</i> , Julian Herold, main supervisor: Prof. Dr. Alexander Schug |
| 07/2018 – 12/2018 | > Analysis and Prediction of Ressource Usage and User Behavior in Scientific Clouds , <i>Karlsruhe Institute of Technology</i> , Sven Lange, main supervisor: Prof. Dr. Achim Streit. |
| 06/2015 – 11/2015 | > Design and Evaluation of an SVM Framework for Scientific Data Applications , <i>Forschungszentrum Jülich</i> , Philipp Glock, main supervisor: Prof. Dr. Morris Riedel |

Bachelor Thesis

- | | |
|-------------------|--|
| 10/2023 – 04/2024 | > Investigation and development of a predictive step response algorithm for the ultra-precise high-voltage regulation of the KATRIN experiment , <i>Karlsruhe Institute of Technology</i> , Fynn Tschacher, main supervisor: Prof. Dr. Kathrin Valerius |
| 03/2023 – ... | > Massively Parallel CMA-ES , <i>Karlsruhe Institute of Technology</i> , Jonathan Roth, main supervisor: Prof. Dr. Achim Streit |
| | > Massively Parallel Particle Swarm Optimization , <i>Karlsruhe Institute of Technology</i> , Paul Zanner, main supervisor: Prof. Dr. Achim Streit |
| 11/2022 – ... | > Random Sampling and Shuffling in Parallel Neural Network Training , <i>Karlsruhe Institute of Technology</i> , Brandon Schühlein, main supervisor: Prof. Dr. Achim Streit |

Supervision (cont.)

- 01/2021 – 07/2021 > **Prediction of Optimal Solvers for Sparse Linear Systems using Deep Learning**, *Karlsruhe Institute of Technology*, Yannick Funck, main supervisor: Prof. Dr. Achim Streit
- 05/2020 – 09/2020 > **Skalierbare Klassifikation von Remote Sensing Daten**, *Karlsruhe Institute of Technology*, Jakob Ohm, main supervisor: Prof. Dr. Achim Streit
- 07/2016 – 12/2016 > **Design and Implementation of a Parallel Artificial Neural Networks**, *Forschungszentrum Jülich*, Matthias Richerzhagen, main supervisor: Prof. Dr. Morris Riedel

Summer Students

- 08/2017 – 10/2017 > **Deep Learning in Hyperspectral Remote Sensing**, *Forschungszentrum Jülich*, Julius Lange
- 08/2016 – 10/2016 > **Cortex Segmentation using CNNs with Spark**, *Forschungszentrum Jülich*, Monika Bajcer

Professional Appointments

Committees

- 2024 – . . . > **European Conference on Artificial Intelligence (ECAI)**, Program Committee.
- 2022 – . . . > **AI HERO Hackathon on Energy-Efficient AI**, Program Committee
- > **H.AI Con**, Program Committee
- 2021 – . . . > **International Supercomputing Conference (ISC)**, Workshops Committee
- 2019 – . . . > **Euro-Par**, Program Committee
- > **Workshop On Scalable Data Analytics In Scientific Computing (SDASC)**, Local Committee Organizers
- 2019 – 2021 > **DAMe Workshop**, Program Committee

Reviewer

- 2024 – . . . > **Nature Machine Intelligence (Nat MIntl)**.
- > **Association for the Advancement of Artificial Intelligence (AAAI)**.
- 2023 – . . . > **European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD)**
- 2022 – . . . > **International Conference on Learning Representations (ICLR)**.
- > **Frontiers in Applied Mathematics and Statistics**
- > **Neural Information Processing Systems (NeurIPS)**
- 2021 – . . . > **International Conference on Machine Learning (ICML)**
- 2020 – . . . > **IEEE Access**
- > **Journal of Grid Computing (GRID)**
- 2019 – . . . > **International Conference on Computational Science (ICCS)**
- > **Journal of Supercomputing (SUPE)**
- > **Workshop On Scalable Data Analytics In Scientific Computing (SDASC)**
- 2019 > **PLOS Computational Biology**
- > **Platform on Advanced Computing (PASC'19)**

Professional Appointments (cont.)

- 2018 > Heterogenity in Computing Workshop (HCW'19)
- > International Supercomputing Conference (ISC)
- 2016 – . . . > Springer Journal on Big Data (JBD)

Academic Self-Government

- 2022 > Professor Appointment Committee “Exascale Algorithms”
Karlsruhe Institute für Technology
- > Helmholtz Incubator Member
- 2021 – . . . > Helmholtz Open Science Office – AI Spokesperson
- 2020 – . . . > HAICORE (Helmholtz AI Compute Resources) Access Board
- 2019 – . . . > Helmholtz AI Coordinator Karlsruhe
- 2019 > Professor Appointment Committee “Machine Learning”,
Karlsruhe Institute für Technology
- 2016 – . . . > NIC Compute Time Committee, *John von Neumann Institut für Computing*

Languages

- > English: C2
- > Polish: B1
- > Swedish: B1
- > French: B1